

DevOps Essentials – Hands-On Workshop

Type: Hands-On Technical Workshop

Target Audience: AI/ML Engineers, Java Developers, Python Developers

Mode: Practical Implementation + Live Demonstrations + Guided Lab

Workshop Overview

This workshop introduces participants to essential DevOps practices and tools used in modern software development. The focus is on implementing CI/CD pipelines, containerizing applications using Docker, and automating deployments using GitHub Actions. The workshop is designed to support AI/ML, Java, and Python development workflows.

Workshop Objectives

By the end of this workshop, participants will be able to:

- Understand CI/CD concepts and the Build-Test-Deploy workflow
- Use Git and GitHub for collaboration using branches and pull requests
- Containerize applications using Docker
- Create GitHub Actions workflows using YAML
- Automate deployment of applications with zero manual intervention
- Implement rollback strategies for failed deployments

Key Topics Covered

1. CI/CD Concepts

- Introduction to Continuous Integration and Continuous Deployment
- Build-Test-Deploy automation lifecycle
- Benefits of CI/CD for AI/ML, Java, and Python projects
- CI/CD pipeline flow and best practices

2. Git and GitHub Collaboration

- Git basics (clone, commit, push, pull)
- Branching strategies for team development

- Pull Requests and code review workflow
- Merge strategies and version control best practices

3. Docker Fundamentals

- What is containerization and why it matters
- Docker images vs containers
- Writing a Dockerfile
- Building and running containers
- Port mapping and environment consistency
- Docker best practices for AI/ML APIs, Java microservices, and Python services

4. GitHub Actions for CI/CD Automation

- Introduction to GitHub Actions
- Understanding workflows, jobs, steps, and runners
- YAML workflow structure
- Automating unit testing on every push
- Automating Docker build process
- Automating deployment triggers

Hands-On Use Case: Automated Deployment of API Service

Use Case Description

Participants will build an end-to-end automated pipeline that packages an API service into a Docker container and deploys it automatically after code is pushed to GitHub. This use case can be implemented using an AI/ML Resume Screening API, a Python REST API, or a Java Spring Boot service.

User Goal

To reduce the time between “Code Complete” and “Live in Production” by implementing a fully automated CI/CD pipeline.

Main Workflow

1. Developer pushes code to GitHub repository
2. GitHub Actions automatically runs unit tests
3. Docker image is built and tagged
4. Docker image is deployed to the target server
5. Application is restarted and made live

Alternative Workflow

- If unit tests fail, the deployment process is stopped automatically
- Developer receives an instant failure notification through GitHub Actions

Exception Handling / Rollback Strategy

- If the new container fails to start, the system automatically rolls back to the previous stable version
- Deployment is marked as failed and logged for troubleshooting

Expected Outcome

At the end of the workshop, participants will have a working CI/CD pipeline that:

- Runs automated tests on every push
- Builds Docker images automatically
- Deploys the service automatically to a server
- Performs rollback in case of failure
- Requires zero manual deployment intervention

Tools and Technologies Used

- Git and GitHub
- Docker
- GitHub Actions (YAML Workflows)
- Sample API Project (AI/ML / Java / Python based)
- Linux Deployment Environment / Server

Workshop Deliverables

Participants will receive:

- Dockerized application project
- GitHub Actions CI workflow file
- GitHub Actions deployment workflow file
- Automated rollback implementation template
- Ready-to-reuse DevOps pipeline template for real projects